CS 215: Task 07: Due: 15-Oct; time as announced on Piazza. No extensions.

1. Fill volumes of a bottling machine are $\mathcal{N}(\mu, \sigma^2)$ with $\sigma = 4$ ml. A random sample of $n = 16$ bottles gives $\bar{x} = 502$ ml.

   (a) If samples of 16 bottles were repeatedly drawn in this way, find an interval around the observed sample mean 502 ml such that the corresponding interval would contain μ in approximately 95% of repeated samples.

   Solution: Assume the bottle fill volumes are independent and identically distributed (i.i.d.) normal random variables, with known population standard deviation $\sigma = 4$ ml. Since the population standard deviation is known, the 95% confidence interval for μ is

   $$\bar{x} \pm z_{0.025}\frac{\sigma}{\sqrt{n}}.$$

   Substituting $\bar{x} = 502$, $z_{0.025} = 1.96$, $\sigma = 4$, and $n = 16$,

   $$502 \pm 1.96\frac{4}{\sqrt{16}} = 502 \pm 1.96.$$

   Thus, the required interval is

   $$\boxed{(500.04,\ 503.96) \text{ ml.}}$$

   In repeated sampling, approximately 95% of intervals constructed by this procedure would contain the fixed population mean μ.

   (b) With the conditions as in part (a), find the smallest sample size for which the margin of error (half-width of the interval) is at most 1 ml.

   Solution: For a known population standard deviation, the margin of error is

   $$E = z_{0.025}\frac{\sigma}{\sqrt{n}}.$$

   We require $E \leq 1$ ml. Therefore,

   $$1.96\frac{4}{\sqrt{n}} \leq 1 \quad \Longrightarrow \quad \sqrt{n} \geq 7.84 \quad \Longrightarrow \quad n \geq 7.84^2 = 61.4656.$$

   The sample size must be an integer, and rounding down would violate the required bound. Hence,

   $$\boxed{n = 62.}$$

   (c) How many bottles should we sample if we want the error to be at most 0.5 ml instead?

   Solution: The margin of error is $E = 1.96(4)/\sqrt{n}$. Requiring $E \leq 0.5$ gives

   $$1.96\frac{4}{\sqrt{n}} \leq 0.5 \quad \Longrightarrow \quad n \geq \left(\frac{1.96(4)}{0.5}\right)^2 = 245.8624.$$

   Rounding up to the next integer,

   $$\boxed{n = 246.}$$

   This is four times the approximate sample size needed for a 1 ml margin of error, because halving the margin of error requires multiplying the sample size by $2^2 = 4$.

(d) Suppose the interval obtained in part (a) is written as

$$502 \pm \delta.$$

If a random sample of 100 individual bottles is taken, would you expect about 95 of them to have fill volumes within

$$502 \pm \delta?$$

Explain your answer.

Solution: No. The interval $502 \pm \delta$ is a confidence interval for the population mean μ; it describes uncertainty in the estimate of the mean, not the spread of individual bottle fills. Individual fills have standard deviation 4 ml.

For an individual bottle $X \sim \mathcal{N}(\mu, 4^2)$, the central 95% of the population lies approximately within

$$\mu \pm 1.96(4).$$

Thus, the interval containing approximately 95% of individual bottle volumes is

$$\boxed{(\mu - 7.84,\ \mu + 7.84) \text{ ml.}}$$

The distinction is that the standard error of the sample mean is $\sigma/\sqrt{n}$, whereas the standard deviation of an individual observation is σ.

2. Your supervisor asks you to estimate the mean running time of a piece of code. You run the program independently, and obtain the following running times (in ms):

$$12,\ 15,\ 11,\ 14,\ 13,\ 16,\ 10,\ 14,\ 12.$$

Assume that the running times are normally distributed.

(a) How would you report the mean running time to your supervisor? Use a 95% confidence level.

Solution: Assume the nine running times are an independent sample from a normal population. Since the population standard deviation is unknown, use a t-interval with $n - 1 = 8$ degrees of freedom.

The sample mean is

$$\bar{x} = \frac{12 + 15 + 11 + 14 + 13 + 16 + 10 + 14 + 12}{9} = \frac{117}{9} = 13 \text{ ms.}$$

The deviations from the mean are

$$-1,\ 2,\ -2,\ 1,\ 0,\ 3,\ -3,\ 1,\ -1,$$

whose squared values sum to 30. Hence,

$$s^2 = \frac{30}{9-1} = 3.75, \qquad s = \sqrt{3.75} \approx 1.9365, \qquad \frac{s}{\sqrt{9}} \approx 0.6455.$$

Using $t_{0.025,8} = 2.306$, the 95% confidence interval is

$$\bar{x} \pm t_{0.025,8}\frac{s}{\sqrt{n}} = 13 \pm 2.306(0.6455) = 13 \pm 1.489.$$

Therefore,

$$\boxed{(11.51,\ 14.49) \text{ ms.}}$$

(b) Your assistant wrongly uses 1.96 in place of the t critical value. Compute the assistant's reported interval and explain the consequence for its actual coverage.

Solution: Using 1.96 in place of the appropriate t-critical value gives

$$13 \pm 1.96(0.6455) = 13 \pm 1.265,$$

so the assistant reports

$$\boxed{(11.73,\ 14.27)\text{ ms.}}$$

This interval is narrower than the correct 95% t-interval. The reason is that the population standard deviation is unknown and has been estimated by s; the t-distribution accounts for the additional uncertainty in this estimate and therefore has a larger critical value than the standard normal distribution. Using 1.96 here consequently gives actual coverage below 95%.

(c) A second assistant constructed a 99% confidence interval. Compare its width to the one in part (a).

Solution: For a 99% confidence interval with 8 degrees of freedom, use $t_{0.005,8} = 3.355$. The standard error is unchanged from part (a), so

$$13 \pm 3.355(0.6455) = 13 \pm 2.166.$$

Thus, the interval is

$$\boxed{(10.83,\ 15.17)\text{ ms.}}$$

It is wider than the 95% interval because a higher confidence level requires a larger critical value, while the sample size and standard error remain the same.

3. Let $Z \sim \mathcal{N}(0,1)$.

(a) Show that

$$M_{Z^2}(t) = E\left[e^{tZ^2}\right] = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.$$

Solution: The standard normal density is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Therefore,

$$\begin{aligned} M_{Z^2}(t) &= E[e^{tZ^2}] \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz^2} e^{-z^2/2}\, dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1-2t)z^2/2}\, dz. \end{aligned}$$

For this integral to be finite, we need $1-2t > 0$, or $t < 1/2$. Let $u = \sqrt{1-2t}\, z$. Then $dz = du/\sqrt{1-2t}$, and

$$M_{Z^2}(t) = \frac{1}{\sqrt{1-2t}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2}\, du = (1-2t)^{-1/2}.$$

Hence

$$\boxed{M_{Z^2}(t) = (1-2t)^{-1/2}, \qquad t < \frac{1}{2}.}$$

(b) Let $Z_1, \ldots, Z_k$ be i.i.d. $\mathcal{N}(0,1)$ and $W = \sum_{i=1}^{k} Z_i^2$. Find $M_W(t)$.

Solution: Since the Z_i's are independent, the random variables Z_i^2 are also independent. The MGF of a sum of independent random variables is the product of their MGFs. Thus,

$$M_W(t) = E\left[e^{t\sum_{i=1}^{k} Z_i^2}\right] = \prod_{i=1}^{k} E[e^{tZ_i^2}]$$
$$= \prod_{i=1}^{k} (1-2t)^{-1/2} = (1-2t)^{-k/2}, \qquad t < \frac{1}{2}.$$

Therefore,

$$\boxed{M_W(t) = (1-2t)^{-k/2}, \qquad t < \frac{1}{2}.}$$

(c) Using $M_W(t)$, show that $E[W] = k$ and $\mathrm{Var}(W) = 2k$.

Solution: From part (b),

$$M_W(t) = (1-2t)^{-k/2}.$$

Differentiating,

$$M_W'(t) = k(1-2t)^{-k/2-1},$$

and

$$M_W''(t) = k(k+2)(1-2t)^{-k/2-2}.$$

The first two moments are obtained by evaluating these derivatives at $t = 0$:

$$E[W] = M_W'(0) = k, \qquad E[W^2] = M_W''(0) = k(k+2).$$

Consequently,

$$\mathrm{Var}(W) = E[W^2] - (E[W])^2 = k(k+2) - k^2 = 2k.$$

Hence,

$$\boxed{E[W] = k, \qquad \mathrm{Var}(W) = 2k.}$$

(d) If $W_1 \sim \chi_a^2$ and $W_2 \sim \chi_b^2$ are independent, what is the distribution of $W_1 + W_2$? Justify your answer.

Solution: Because W_1 and W_2 are independent, the MGF of their sum is the product of their MGFs. Therefore,

$$M_{W_1+W_2}(t) = M_{W_1}(t) M_{W_2}(t)$$
$$= (1-2t)^{-a/2}(1-2t)^{-b/2}$$
$$= (1-2t)^{-(a+b)/2}.$$

This is the MGF of a chi-square random variable with $a+b$ degrees of freedom. Since MGFs uniquely determine distributions when they exist in a neighbourhood of zero,

$$\boxed{W_1 + W_2 \sim \chi_{a+b}^2.}$$

4. Let $X_1, \ldots, X_{10}$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ and let S^2 be the sample variance.

(a) Find $E[S^2]$, $\mathrm{Var}(S^2)$ and the standard deviation of S^2 expressed as a multiple of σ^2.

Solution: Here $n = 10$. By the stated chi-square result,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

so we can write $S^2 = (\sigma^2/9)W$, where $W \sim \chi^2_9$. A chi-square variable with 9 degrees of freedom has mean 9 and variance 18. Hence,

$$E[S^2] = \frac{\sigma^2}{9}E[W] = \frac{\sigma^2}{9}(9) = \sigma^2,$$

and

$$\mathrm{Var}(S^2) = \frac{\sigma^4}{9^2}\,\mathrm{Var}(W) = \frac{18\sigma^4}{81} = \frac{2\sigma^4}{9}.$$

Taking the square root,

$$\boxed{\mathrm{sd}(S^2) = \frac{\sqrt{2}}{3}\sigma^2 \approx 0.471\,\sigma^2.}$$

(b) Find constants c_1 and c_2 such that

$$P(S^2 < c_1\sigma^2) = 0.05, \qquad P(S^2 > c_2\sigma^2) = 0.05.$$

Solution: Let $W = 9S^2/\sigma^2$, so $W \sim \chi^2_9$. For the upper-tail probability,

$$P(S^2 > c_2\sigma^2) = P(W > 9c_2) = 0.05.$$

By the given chi-square critical value, $9c_2 = \chi^2_{0.05,9} = 16.919$, giving

$$c_2 = \frac{16.919}{9} \approx 1.880.$$

Similarly,

$$P(S^2 < c_1\sigma^2) = P(W < 9c_1) = 0.05.$$

Since $\chi^2_{0.95,9} = 3.325$ is the lower 5th percentile under the stated upper-tail convention,

$$9c_1 = 3.325 \quad \Longrightarrow \quad c_1 = \frac{3.325}{9} \approx 0.369.$$

Thus,

$$\boxed{c_1 \approx 0.369, \qquad c_2 \approx 1.880.}$$

5. A random sample of $n = 11$ resistors from a normal population gives the sample standard deviation $s = 0.8\ \Omega$.

(a) Construct a 95% confidence interval for σ^2.

Solution: Assume the resistor resistances are an i.i.d. sample from a normal population. For a normal population,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Here $n = 11$, $s = 0.8\ \Omega$, and $(n-1)s^2 = 10(0.8)^2 = 6.4$. Using the supplied critical values,

$$P\left(3.247 \le \frac{10S^2}{\sigma^2} \le 20.483\right) = 0.95.$$

To isolate σ^2, invert the inequalities. Since all quantities are positive, inversion reverses the order:

$$P\left(\frac{6.4}{20.483} \le \sigma^2 \le \frac{6.4}{3.247}\right) = 0.95.$$

The 95% confidence interval is therefore

$$\boxed{(0.312,\ 1.971)\ \Omega^2.}$$

(b) Construct a 95% confidence interval for σ.

Solution: Since $\sigma > 0$, take the square root of each endpoint of the variance interval from part (a):

$$\left(\sqrt{0.312},\ \sqrt{1.971}\right) \approx \boxed{(0.559,\ 1.404)\ \Omega.}$$

(c) Explain why the interval in part (a) is not symmetric about s^2.

Solution: The interval is obtained by transforming two chi-square critical values. The chi-square distribution is right-skewed, so its lower and upper critical values are not equally spaced around its mean (or around $n-1$). Moreover, solving the pivotal inequality for σ^2 puts the larger chi-square critical value in the denominator of the lower endpoint and the smaller one in the denominator of the upper endpoint. The resulting interval is therefore not symmetric about $s^2 = 0.64\ \Omega^2$.

(d) The specification requires $\sigma \le 1.5\ \Omega$. Based on your interval, what can you conclude at the 95% confidence level?

Solution: The 95% confidence interval for σ is $(0.559,\ 1.404)\ \Omega$. Its upper endpoint is below the specification limit $1.5\ \Omega$. Thus the entire confidence interval lies within the acceptable range, and we can conclude at the 95% confidence level that the population standard deviation satisfies

$$\boxed{\sigma < 1.5\ \Omega.}$$

6. Let $X_1, \ldots, X_5$ be i.i.d. $\mathcal{N}(0, \sigma^2)$ and define

$$Y = c\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.$$

(a) Find the constant $c > 0$ for which Y has a student's t distribution, and state its degrees of freedom. c should not depend on σ.

Solution: The numerator is normal:

$$X_1 + X_2 \sim \mathcal{N}(0, 2\sigma^2), \qquad Z = \frac{X_1 + X_2}{\sigma\sqrt{2}} \sim \mathcal{N}(0,1).$$

For the denominator, each X_i/σ is standard normal, so

$$V = \frac{X_3^2 + X_4^2 + X_5^2}{\sigma^2} \sim \chi^2_3.$$

By the definition of the Student t-distribution, if $Z \sim \mathcal{N}(0,1)$, $V \sim \chi^2_\nu$, and Z, V are independent, then $Z/\sqrt{V/\nu} \sim t_\nu$. Here $\nu = 3$, and

$$\begin{aligned}\frac{Z}{\sqrt{V/3}} &= \frac{X_1 + X_2}{\sigma\sqrt{2}}\sqrt{\frac{3\sigma^2}{X_3^2 + X_4^2 + X_5^2}} \\ &= \sqrt{\frac{3}{2}}\,\frac{X_1 + X_2}{\sqrt{X_3^2 + X_4^2 + X_5^2}}.\end{aligned}$$

Comparing this with the definition of Y, we obtain

$$\boxed{c = \sqrt{\frac{3}{2}} \approx 1.225, \qquad Y \sim t_3.}$$

The factor σ cancels, as required.

(b) State which independence property you used in your construction.

Solution: The variable Z is a function only of X_1, X_2, while V is a function only of X_3, X_4, X_5. These two groups of observations are independent because all five X_i's are independent. Functions of independent groups of random variables remain independent. Therefore,

$$\boxed{Z \text{ and } V \text{ are independent.}}$$